

Appendix 8: How to Parse and Crack the In-Vehicle CAN Protocol - DBC Protocol

Instruction Manual

Instruction Manual Version: V2.04

Update Date: 2023.07.01

DBC Protocol Parsing

Each CAN channel can independently support the parsing of vehicle CAN-bus application protocols, as long as the user imports the relevant DBC file in CANtest or CANpro1.50 software, the parsing of application layer data can be realized. It can be used for vehicle CAN protocol analysis, vehicle testing and maintenance, and cracking vehicle CAN protocol.

When users use CANTest or CANPro1.50, they only need to replace ControlCAN.dll and other library files (refer to: How to Compatible with the use of Zhou Ligong CANTest/CANpro1.50 software.pdf), and select the model number: USBCAN-2E-U can be.

CANpro1.50 is more feature-rich, here take CANpro1.50 as an example.

1、Operation Steps

Open the CANpro1.50 software, select the USBCAN-2E-U interface card, and select the baud rate of the bus (subject to the actual baud rate, the automotive CAN bus is generally 500K), click OK and start to start the CAN interface card. As shown in Figure 2;

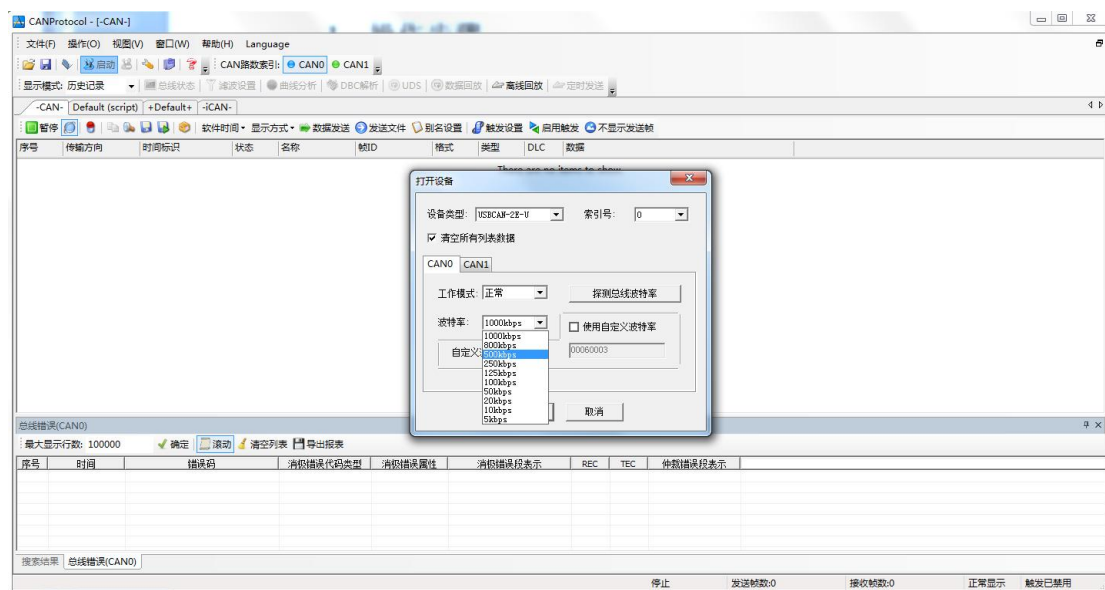


Figure 2 Open CANpro Software Initialization

Click the DBC Parsing button in the menu shortcut operation to enter the DBC Parsing interface, as shown in Figure 3;

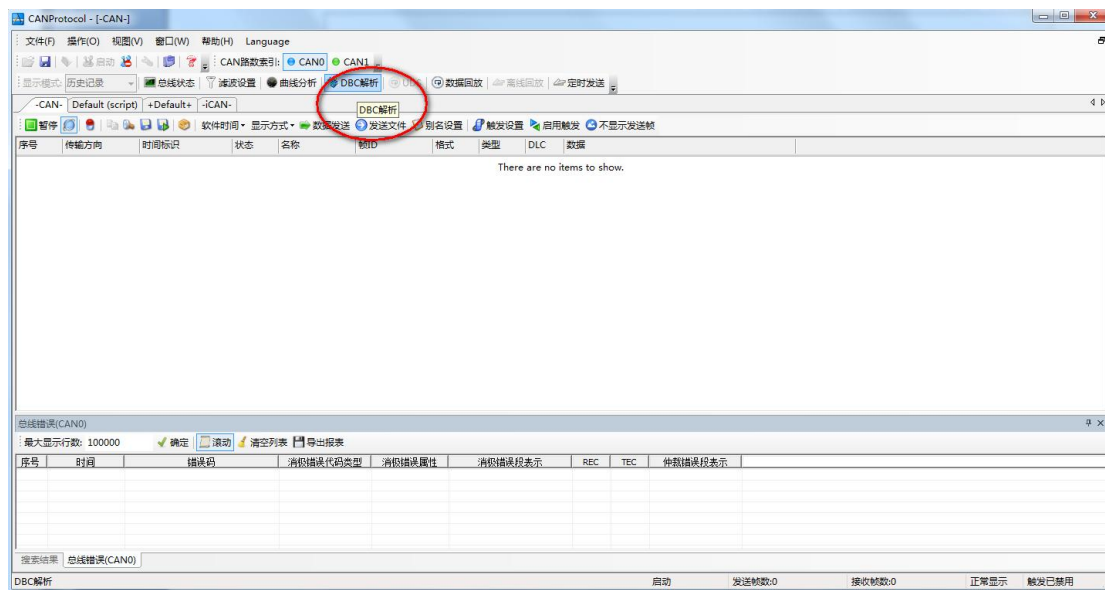


Figure 3 Open DBC Parsing

In the DBC parsing interface, click Load DBC and select the corresponding DBC file to open, CANpro comes with three DBC files. In this paper, take J1939 protocol as an example, select j1939.dbc to open it and parse the protocols of diesel engine, truck or bus, etc., as shown in Figure 4.

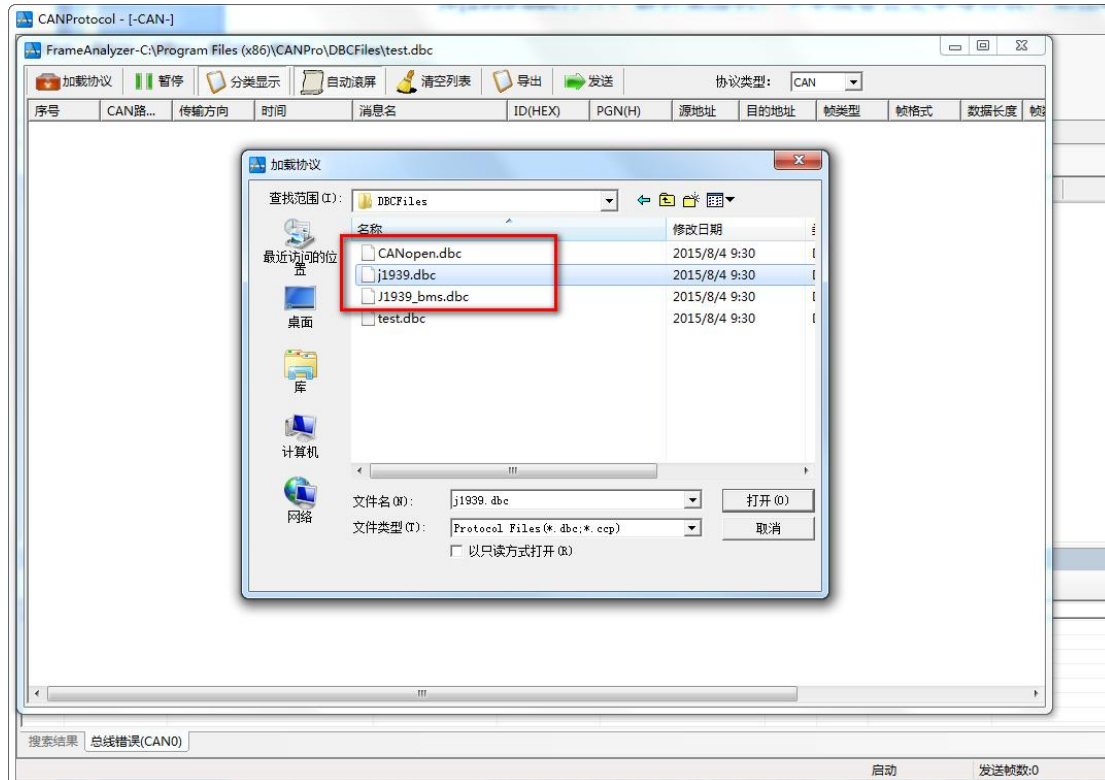


Figure 4 Load DBC File

At this point, the received data can be parsed by DBC, and the user can view it by sorting or refreshing the display. Click on a message, the application data contained in this frame will be displayed in the parsing box below. As shown in Figure 5, ID 0x0CF0041A in the fourth byte is 0x6C, the fifth byte is 0xD6. check, against the SAE_J1939-71 protocol: Electronic Engine Controller #1: EEC1 (message name) in the fourth and fifth bytes represent EngSpeed (engine speed).

Data length: 2 bytes

Resolution: 0.125 rpm/bit increments from 0 rpm (high byte resolution = 32 rpm/bit)

Data range: 0 to 8031.875 rpm

RPM can be calculated: $0xD66C * 0.125$ is 6861.50 rpm (revolutions per minute).

For the definition and resolution of other parameters, please refer to SAE_J1939-71 Protocol: CD-ROM\Instructions Documentation Table of Contents\16.Annex 9: SAE_J1939-71 Protocol.pdf

The screenshot shows the FrameAnalyzer software interface. The top window displays a list of received CAN frames with columns for sequence number, direction, time, message name, ID, source address, destination address, data type, data format, data length, and raw data. The bottom window shows a list of signals with columns for sequence number, signal name, actual value, value description, original value, start bit, width, conversion ratio, and conversion offset.

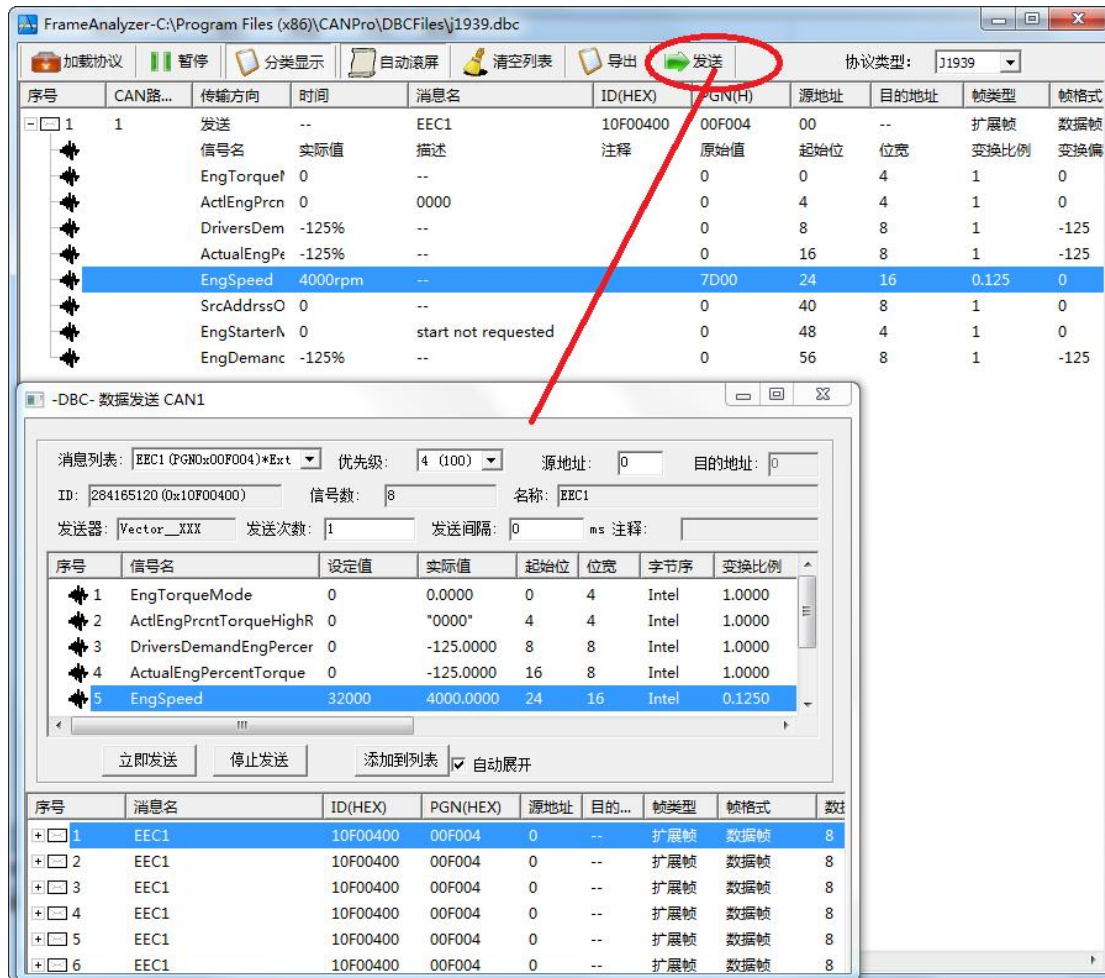
序号	传输方向	时间	消息名	ID	源地址	目的地址	数据类型	数据格式	数据长度	帧数据
0	接收	532.2626	EEC2	00F00302 H	02 H	--	扩展帧	数据帧	8	00 00 30 00 00 00 00
1	接收	532.2489	EEC1	0CF0041A H	1A H	--	扩展帧	数据帧	8	00 00 00 60 D6 00 00 00
2	接收	532.2598	HOURS	00FEE505 H	05 H	--	扩展帧	数据帧	8	D2 09 00 00 00 00 00 00
3	接收	532.2070	ET1	00FEE01 H	01 H	--	扩展帧	数据帧	8	14 14 00 00 00 00 00 00
4	接收	532.2215	VEP1	00FEF704 H	04 H	--	扩展帧	数据帧	8	00 00 00 00 00 00 D0 68
5	接收	532.2267	SHUTDN	00FEE407 H	07 H	--	扩展帧	数据帧	8	00 00 00 00 00 00 00 00
6	接收	532.2422	EFL_P1	00FEFF03 H	03 H	--	扩展帧	数据帧	8	00 00 00 27 00 00 00 00
7	接收	532.2458	IC1	00FEF606 H	06 H	--	扩展帧	数据帧	8	00 00 27 00 00 00 00 00

序号	信号名	实际值	值描述	原始值	起始位	位宽	变换比例	变换偏移
0	EngTorqueMode	0.00	--	0	0	4	1.000000	0.000000
1	ActEngPrntTorqueHighResolution	0.00	0000	0	4	4	1.000000	0.000000
2	DriversDemandEngPercentTorque	-125.00%	--	0	8	8	1.000000	-125.000000
3	ActualEngPercentTorque	-125.00%	--	0	16	8	1.000000	-125.000000
4	EngSpeed	6861.50rpm	--	54892	24	16	0.125000	0.000000
5	SrcAddrssOfCtrlngDvcForEngCtrl	0.00	--	0	40	8	1.000000	0.000000
6	EngStarterMode	0.00	start not requested	0	48	4	1.000000	0.000000
7	EngDemandPercentTorque	-125.00%	--	0	56	8	1.000000	-125.000000

Figure 5 DBC Protocol Analysis Results

Tip: When using the classification display function, the software will mark the data with changes in red, which can help users to quickly complete the identification of variables when deciphering unknown protocols. For example, if you want to know the CANID and data segment corresponding to the steering wheel, you can use this method to run, turn the steering wheel, and observe the variables that turn red, that is, the corresponding.

The latest version of CANPro 1.50 software supports the DBC protocol sending function.

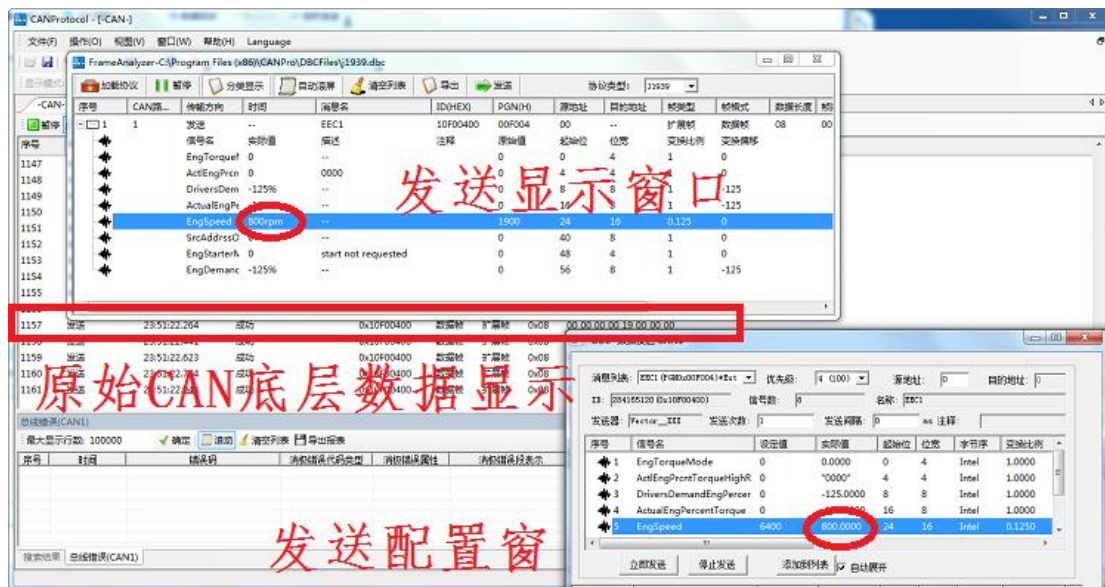
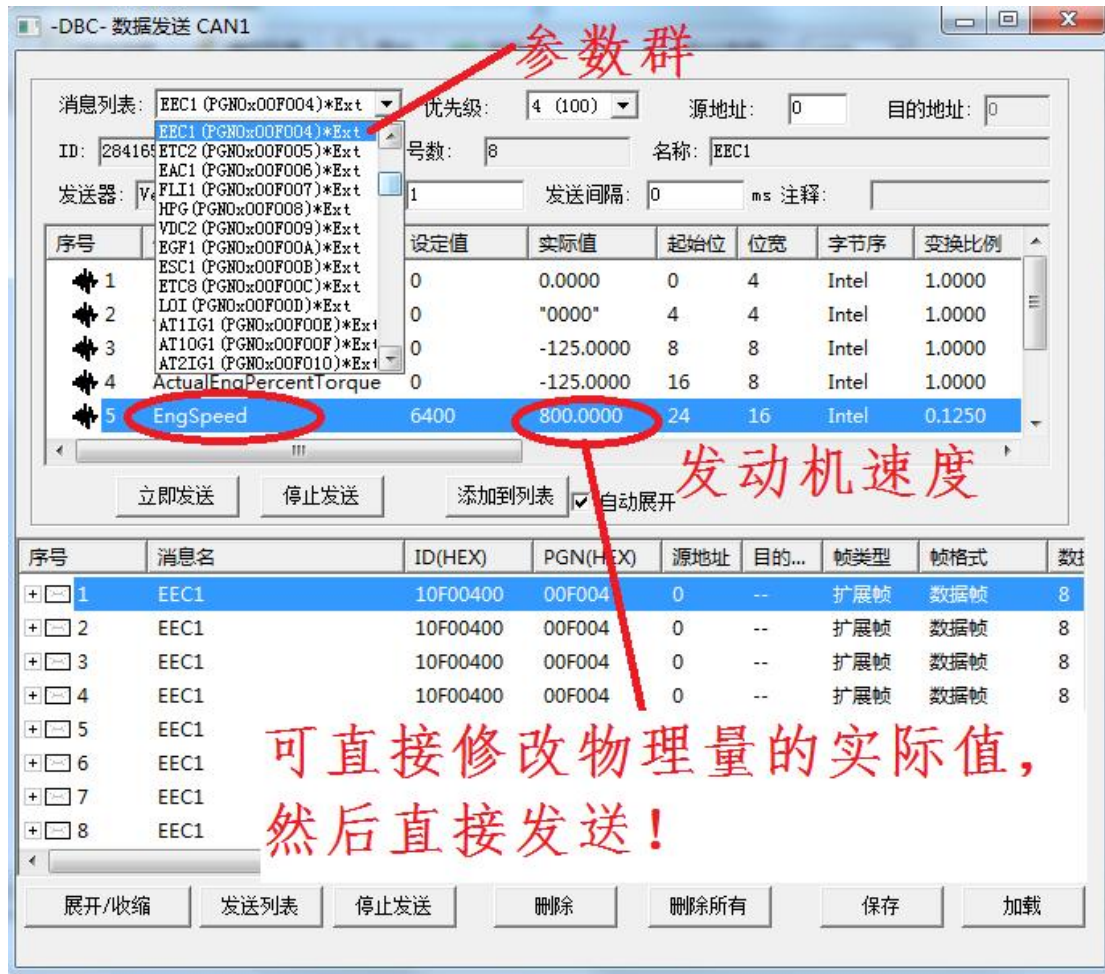


Supports visual configuration of sending data according to J1939 protocol.

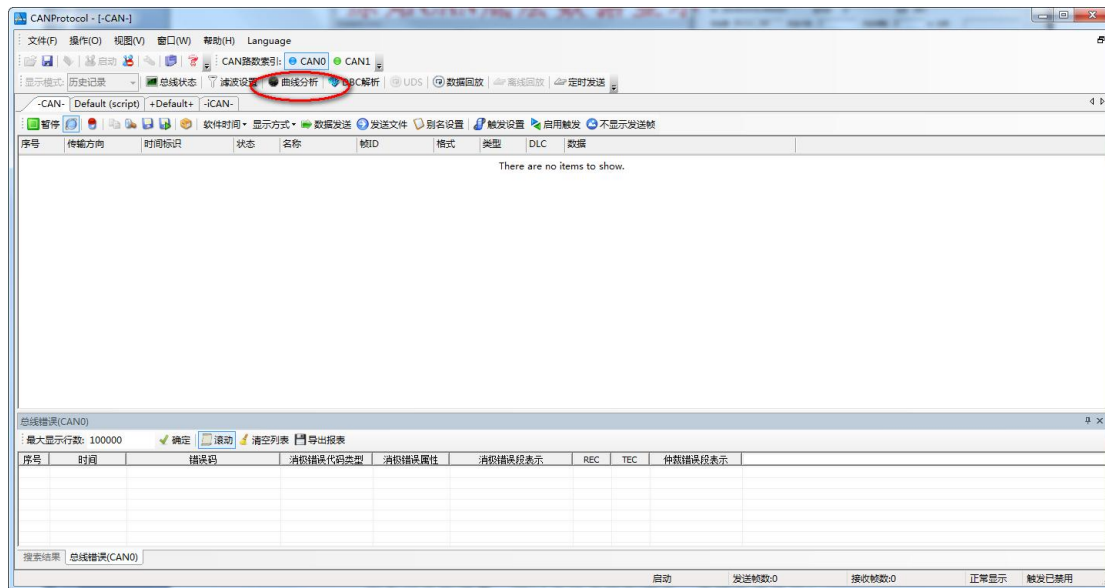
J1939 protocol sending interface:

Any physical parameter of the parameter group is defined with actual value and sent. Ignore the principle of J1939 protocol. You can directly simulate the J1939 device for testing.

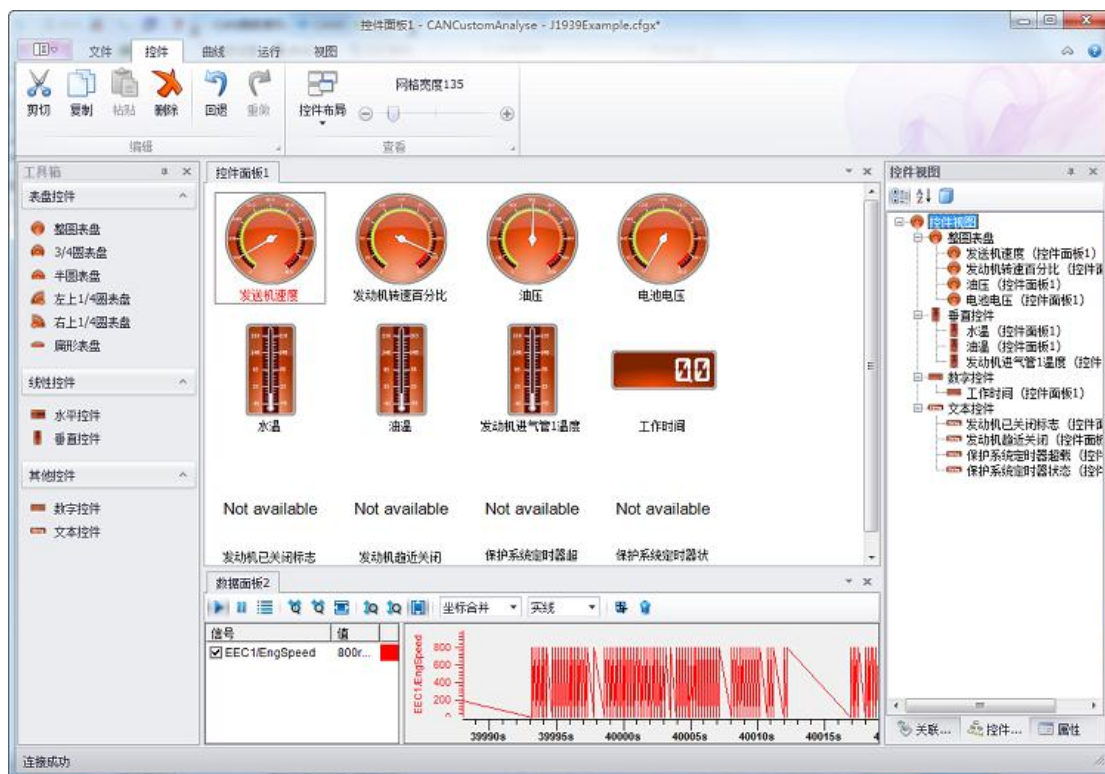
In the following figure, configure the simulated engine speed of 800 rpm and send.



The latest version of CANPro 1.50 software supports DBC protocol curve and graph display.



All relevant physical quantities can be directly associated to graphs and curves for visualization!



If you don't have a real car, you can also debug and learn the J1939 protocol, buy a dual-channel product, and you can send and receive data between the two channels.

CANPro - [-CAN-]

文件(F) 操作(O) 视图(V) 窗口(W) 帮助(H) Language

CAN路数索引: CAN0 CAN1

显示模式: 历史记录 总线状态 滤波设置 曲线分析 DBC解析 UDS 数据回放 离线回放 定时

-CAN- Default (script) +Default+ -iCAN-

暂停 软件时间 显示方式 数据发送

底层CAN数据列表

序号	传输方向	时间标识	状态	名称	数据帧	扩展帧	数据
1193	发送	23:57:11.483	成功	0x10F00400	数据帧	扩展帧	0x08 00 00

数据面板1 - CANCustomAnalyse - 车速.crgx

查看模式 分析模式 水平放大 水平缩小 水平全屏 垂直放大 垂直缩小 垂直全屏

坐标显示 坐标合并 曲线线型 实线 显示内部工具栏 导出图片 清除

工具箱

表盘控件

- 整圆表盘
- 3/4圆表盘
- 半圆表盘
- 左上1/4圆表盘
- 右上1/4圆表盘
- 扁形表盘

线性控件

- 水平控件
- 垂直控件

其他控件

- 数字控件
- 文本控件

控件面板1

发动机转速

数据面板1

信号 值

EEC1/EngSpeed 4000...

EEC1/EngSpeed

40800s 40900s

连接成功

消息列表: 284185120 (0x10F00400) 优先级: 1 源地址: 0 目的地址: 0

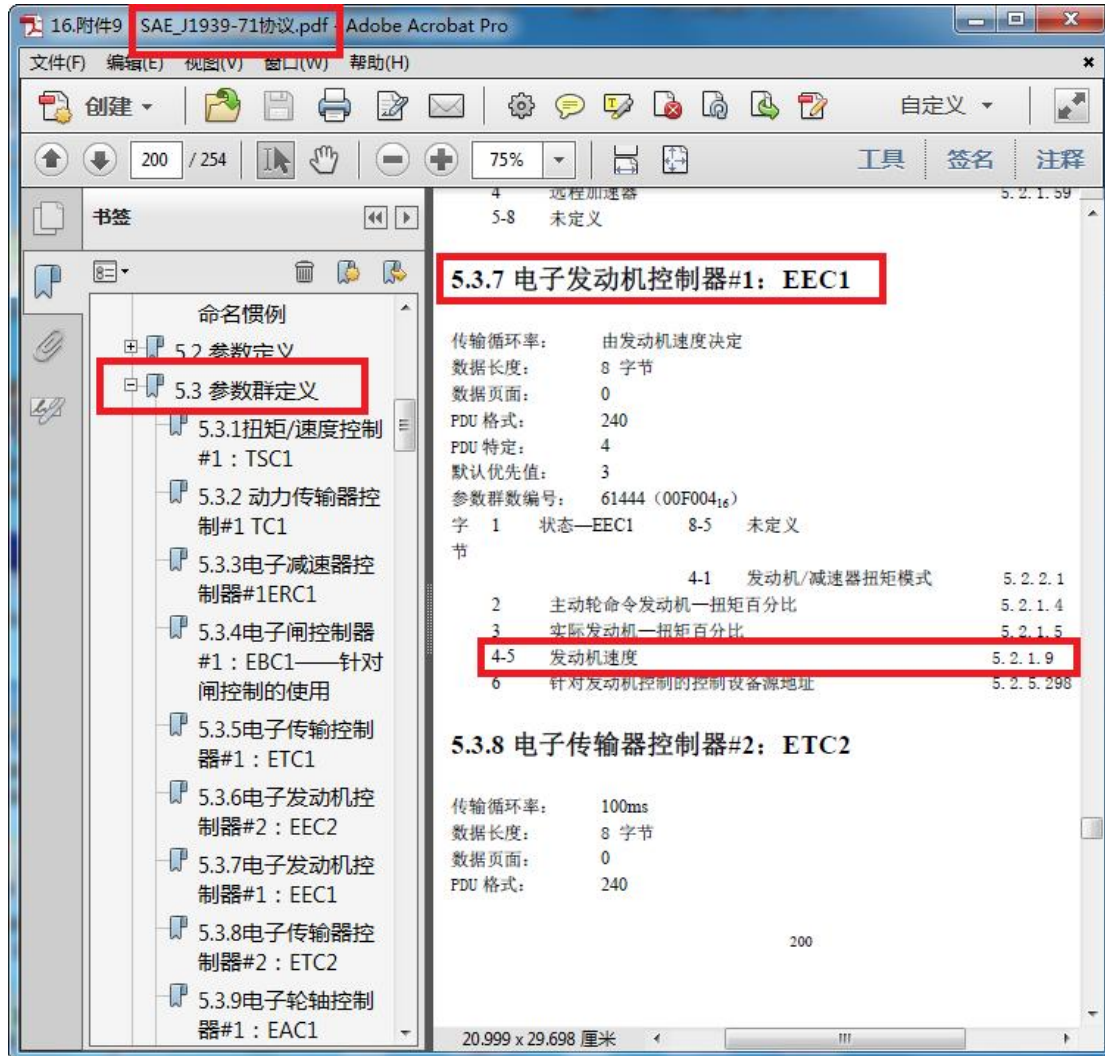
ID: 284185120 (0x10F00400) 信号数: 8

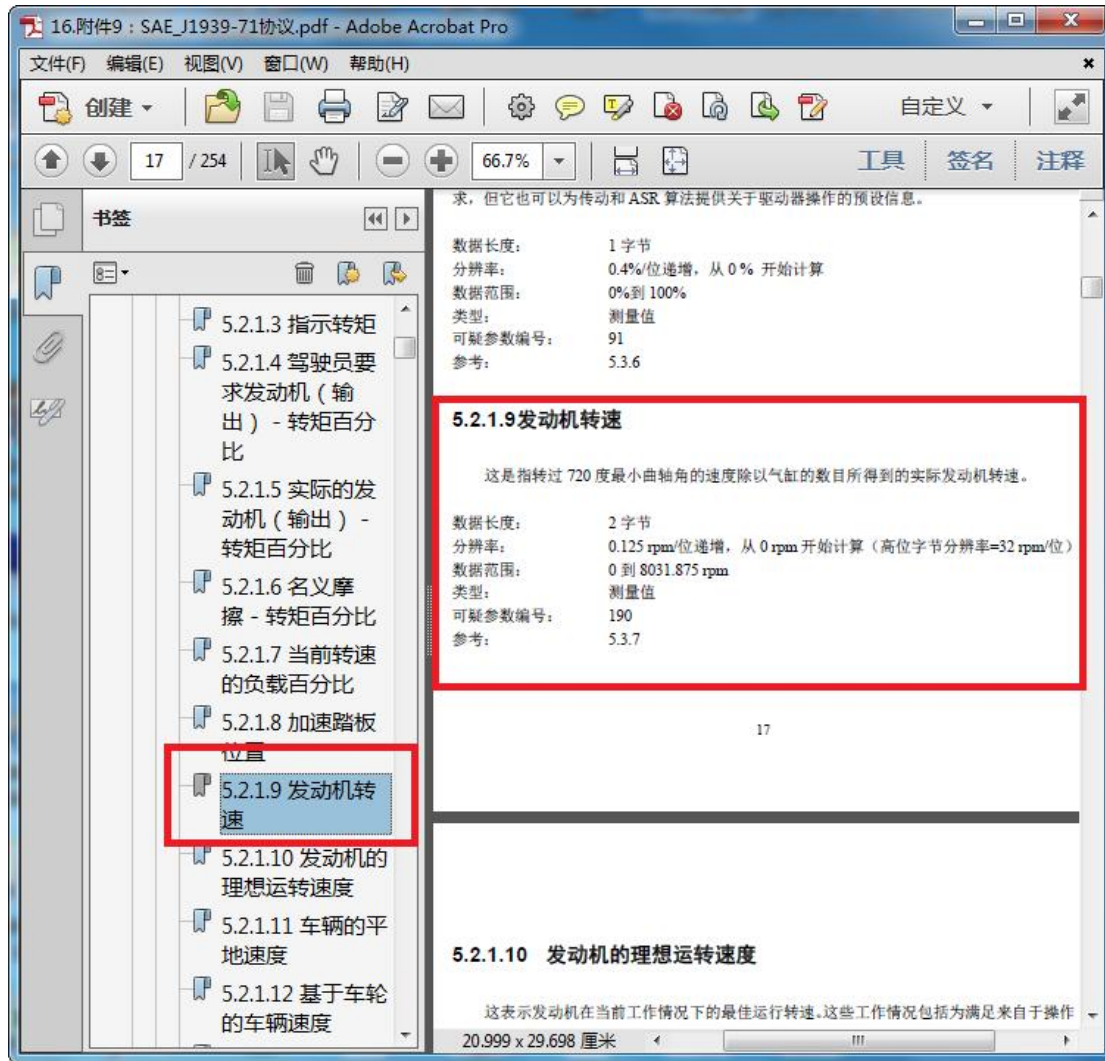
发送器: Vector_XXX 发送次数: 1

CAN1通道按J1939协议发送数据!

序号	信号名	设定值	数据	格式	单位
1	EngTorqueMode	0	0.0000	Intel	1.0000
2	ActEngPrntTorqueHighR	0	"0000"	Intel	1.0000
3	DriversDemandEngPerce	0	-125.0000	Intel	1.0000
4	ActualEngPercentTorque	0	-125.0000	Intel	1.0000
5	EngSpeed	32000	4000.0000	Intel	0.1250

The parameter groups in the software are displayed in English, and the names of the controls in the control panel can be modified to Chinese, and the Chinese version of the protocol is provided here for comparison.





2、Application Scope

- Industrial Control Testing
- Automotive Electronics Maintenance and Repair
- Protocol Cracking